

AST 6007-01 Spacecraft Dynamics and Control

Final Examination

December 14, 2011

Open Book/Open Notes

Work all problems. In the beginning of your answer sheet, sign the honor code: I never received nor gave help regarding this exam.

1. (15) A quaternion vector given by $\left(\frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{2}}{2}\right)$ represents the attitude of a rigid body. Determine the rotation matrix that represents this attitude of the rigid body.
2. (15) A quaternion vector (q_1, q_2, q_3, q_4) satisfies the following rotational kinematic equations

$$\begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \end{bmatrix} = \frac{1}{2} S(\omega) \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix} + \frac{1}{2} q_4 \omega$$
$$\dot{q}_4 = -\frac{1}{2} \omega^T \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix}$$

Show that if the initial values of the quaternion satisfy

$$q_1^2(0) + q_2^2(0) + q_3^2(0) + q_4^2(0) = 1$$

then

$$q_1^2(t) + q_2^2(t) + q_3^2(t) + q_4^2(t) = 1$$

holds for all $t \geq 0$.

3. An inertially symmetric rigid spacecraft has a diagonal inertia matrix given by $I = \text{diag}(200, 200, 200) \text{ kg} \cdot \text{m}^2$. It is desired that the spacecraft perform a rotational rest-to-rest maneuver that transfers the spacecraft from its initial attitude given by a rotational matrix representation $R = \exp[S(a)]$, where the unit vector $a = (0.424, -0.566, 0.707)$, to the final attitude given by the rotational matrix R that is the identity. The rotational maneuver should be a rotation about the fixed axis defined by the unit vector a (it is an Euler rotation), and the rotational acceleration about that fixed axis should be either $+0.01 \text{ rad/sec}^2$ or -0.01 rad/sec^2 . Assume gravity gradient moments can be ignored.
 - a. Give an analytical expression for the time dependence of each component of the spacecraft angular velocity vector expressed in the body-fixed coordinate frame.
 - b. Give an analytical expression for the time dependence of each component of the control moment vector expressed in the body-fixed coordinate frame.

c. How long is a time period required to complete the rotational maneuver?

4. Consider the pitch axis control model for a spacecraft in a circular orbit given by

$$I_y \Delta \ddot{\theta} = 3\omega_0^2 (-I_x + I_z) \Delta \theta + M_y$$

where the orbital velocity vector $\omega_0 = 0.0012 \text{ rad/sec}$, the spacecraft principal moments of inertia are $I = \text{diag}(150, 200, 100) \text{ kg} \cdot \text{m}^2$. In this equation $\Delta \theta$ denotes the pitch angle and M_y denotes the pitch axis control moment. It is desired to perform a rest-to-rest maneuver using the pitch axis torques that transfers the initial pitch angle $\Delta \theta = 45^\circ$ to the final pitch angle $\Delta \theta = 0^\circ$.

It is claimed that this rest-to-rest maneuver can be performed using a single impulse supplied by the pitch axis control moment.

Is this claim true or false?

If the claim is true, then describe at what time instant the impulse should be applied and describe the required change in the pitch rate due to this impulse.

If the claim is false, justify that it is not possible to accomplish the maneuver using a single impulse.

5. Consider the pitch axis control model for a spacecraft in a circular orbit given by

$$I_y \Delta \ddot{\theta} = 3\omega_0^2 (-I_x + I_z) \Delta \theta + M_y$$

where the orbital frequency $\omega_0 = 0.0012 \text{ rad/sec}$, the spacecraft principal moments of inertia are $I = \text{diag}(150, 100, 200) \text{ kg} \cdot \text{m}^2$. In this equation $\Delta \theta$ denotes the pitch angle and M_y denotes the pitch axis control moment.

a. If the pitch axis control moment $M_y = 0$, show that the pitch equilibrium $(\Delta \theta, \Delta \dot{\theta}) = (0, 0)$ is an unstable equilibrium

b. Show that linear state feedback to the pitch axis control moment can be used to stabilize the pitch axis equilibrium $(\Delta \theta, \Delta \dot{\theta}) = (0, 0)$ of the spacecraft. Choose feedback gains so that the equilibrium of the closed loop is asymptotically stable with damping ratio of 0.7 and natural frequency of 0.008 rad/sec (that is, the closed loop characteristic polynomial is $s^2 + 0.0112s + 0.000064$).

$$\begin{bmatrix} \Delta \dot{\theta} \\ \Delta \ddot{\theta} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\omega_0^2 & 0 \end{bmatrix} \begin{bmatrix} \Delta \theta \\ \Delta \dot{\theta} \end{bmatrix}$$

$$\begin{bmatrix} \Delta \theta \\ \Delta \dot{\theta} \end{bmatrix} = e^{At} \begin{bmatrix} \Delta \theta_0 \\ \Delta \dot{\theta}_0 \end{bmatrix} = e^{At} \begin{bmatrix} \pi/4 \\ 0 \end{bmatrix}$$